

SARVESWARA RAO KOPPULA

System Engineer | GoLang Developer

+91–8297116131 | sarru1729@gmail.com | LinkedIn

Aiming to leverage extensive software development skills to create and implement cutting-edge solutions that improve operational processes and contribute to the overall success of a forward-thinking organization, preferably in Hyderabad.

Education

- **2021:** Masters in Computer Science and Engineering from NIT Nagpur (CGPA – 7.40)
- **2018:** Bachelors in Computer Science and Engineering from Bapatla Engineering College, Bapatla, Guntur (CGPA – 8.17)

Areas of Expertise

- Golang
- RESTful Microservices & gRPC
- Golang API Frameworks
- Gin, Gorilla Mux
- Databases: PostgreSQL,
- MongoDB, Redis, Connection Pooling
- Queues and Consumers
- Worker Pools,Kafka, RabbitMQ
- AWS Cloud (EC2, S3, EKS, ECR)
- JWT Authentication, OAuth
- System Architecture
- Scalable System Design
- Containerization
- Docker, Kubernetes (EKS)
- CI/CD: Jenkins, GitLab CI
- Collaboration: Open Project
- Python: Banned keyword ML Project
- Kibana, Grafana, New Relic
- API Documentation: Swagger
- Domain: GST & Financial Project

Technical Skills

- **Languages:** Golang, Java,Python,
- **API Development:** RESTful,gRPC
- **Frameworks:** Gin, Gorilla Mux
- **Databases:**PostgreSQL,MongoDB
- **Messaging Systems:** Kafka, RabbitMQ
- **Cloud & DevOps:** AWS (EC2, S3, EKS, ECR), Docker, K8S, Linux
- **CI/CD Tools:** Jenkins, GitLab
- **Monitor & Logging:** Prometheus, Grafana, Kibana, New Relic
- **Domain:** GST, PAN Compliance, Financial Processing

Academic Achievements

- Ranked 1891 All India in GATE-CSE, showcasing competitive proficiency.
- Secured 1st place in Software Contest 2016 held in Bapatla.

Personal Details

Date of Birth: April 2, 1997
Languages: English, Hindi, Telugu
Current Address: Noida, U.P – 201305
Permanent Address: Guntur-Andhra Pradesh - 522612

Profile Summary

- Possess **over 4+ years of experience in engineering robust applications and services** that adjust to evolving industry demands.
- Working as a **System Engineer at IndiaMart,** leading development of **Golang APIs** with seamless integration across diverse databases.
- Developed strong technical foundations across **languages, databases, and architecture,** focusing primarily on **Golang and PostgreSQL.**
- Directed the **Real-Time GST Verification Project,** reducing processing time while improving system-wide accuracy.
- Built **Python ML pipelines** for conversational **AI** and **RAG** compliance, improving detection precision from 60 → **90%** and recall from 70 → **95%**.
- Received consecutive **Pillar of Success Awards in March & April 2022**, in recognition of consistent excellence and impactful contributions to team goals.

Work Experience

IndiaMart Intermesh Ltd. | Noida **Feb'21 – Present**

Designation: System Engineer

Responsibilities:

- Led design and development of high-performance, concurrent Go microservices and APIs (REST/gRPC), ensuring scalable architecture, optimized resource use, and backward compatibility.
- Building containerized applications with Docker and orchestrating deployments via Kubernetes for auto-scaling and efficient resource utilization.
- Integrating CI/CD pipelines using Jenkins, GitLab CI, or ArgoCD to automate testing, builds, and deployments for faster and reliable delivery.
- Utilizing Kafka, NATS, or RabbitMQ to enable fault-tolerant, decoupled systems handling large-scale asynchronous workloads.
- Implementing real-time monitoring, logging, and alerting using Prometheus, Grafana, ELK stack, and Kibana APM to detect and resolve performance issues.
- Optimizing PostgreSQL-based solutions to ensure data integrity and enhance query performance across applications.
- Documenting system architecture and workflows to support scalable development and long-term maintainability.

Achievements:

- Successfully delivered the Real-Time GST Verification Project using GoLang, which improved processing efficiency by 90%, reducing operational costs.
- Recognized with the Pillar of Success Award for outstanding contributions to project success and team collaboration for the months of March (Link) and April 2022 (Link).
- Enhanced API response times by 75% through optimization techniques, resulting in improved user satisfaction and engagement.

Organizational Projects

- Real-Time GST, TAN & PAN **Verification System** – IndiaMart (Golang, Worker Pools, RabbitMQ)
- Implemented **multi-window rate limiting** (1 min, 5 min, 30 min, 1 hour, 1 day) in **IndiaMart Varnish cache** layer to prevent parsing attacks. Integrated **Google reCAPTCHA** for blocked users.
- Php to GoLang APIs Migration (107 APIs)
- Refactored a **15,000-line PHP** Lead Auto-Approval consumer code into a modular **Golang-Kafka** service, cutting memory usage by **95%**. Integrated **Kibana** for telemetry and logging.
- Scalable URL Shortening Service for Simple and User-friendly Domain
- Dockerized APIs into single-page architecture and deployed on AWS EKS (Kubernetes), resulting in 1 million INR annual cost savings by moving services to EKS and decommissioning all EC2 instances
- Designed and developed **a conversational IndiaMART Help AI Assistant** using the **Google Agent Development Kit**, integrating multiple internal knowledge sources via **MCP servers built on the Anthropic Model Context Protocol** to support large language model (LLM) interactions.
- Built a custom **Retrieval-Augmented Generation (RAG)** system from scratch to determine whether a product or buylead is banned or approved on IndiaMART by leveraging similar items and banned keyword context. **Utilized the Google Text Embedding model for vector generation and Gemini 1.5 Pro for response generation**, significantly improving model performance **(Precision:** 60% → **90%, Recall:** 70% → **95%).**
- Designed a Scalabe Service for Grow (for sending GST / HSN details) using **pub-sub** Architecture